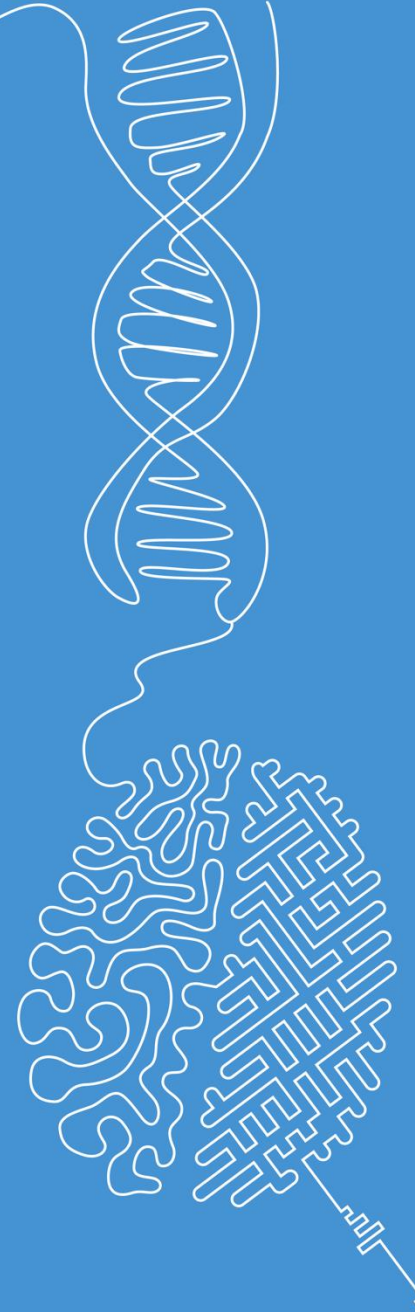


Signals, Sinusoids, and Sound

Image and Signal Processing

Norman Juchler



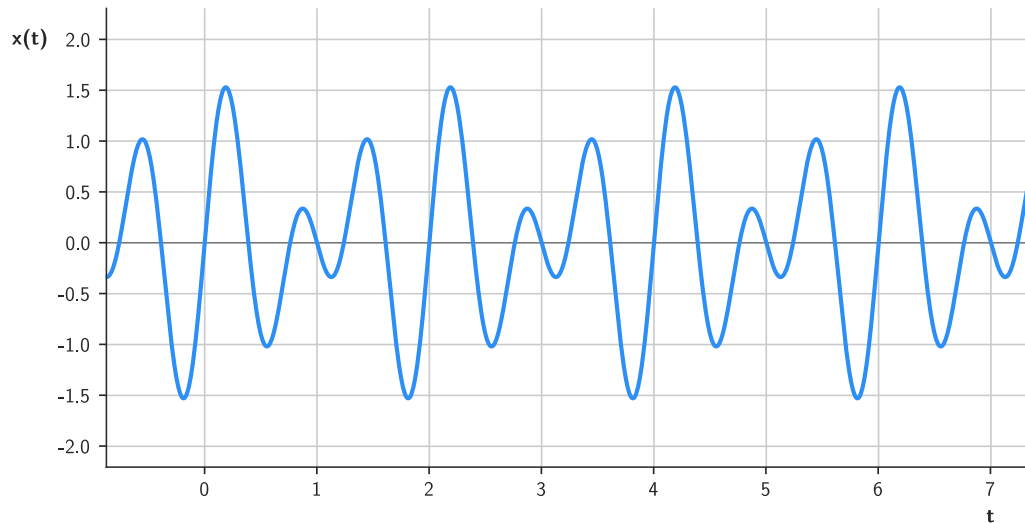
Signals

Mathematical representation and modeling

Signals as mathematical functions

Continuous-time: $x := x(t)$

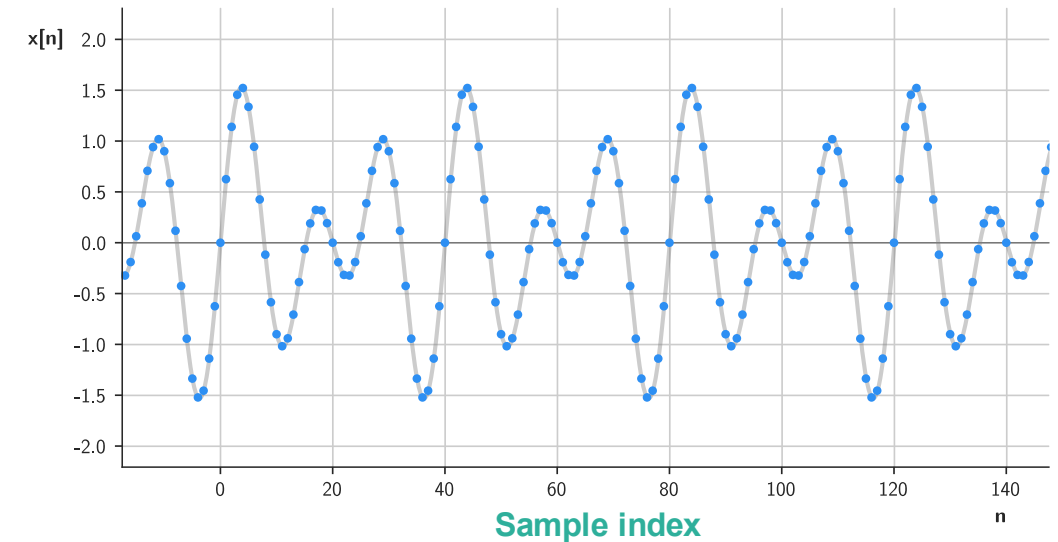
A signal can be represented as a mathematical function that maps a value $x=x(t)$ for each time point t .



- t is the independent variable
- x is the dependent variable

Discrete-time: $x := x[n] = x(n \cdot T_s)$

A signal can also be represented as a discrete 1D sequence which is a function of an index variable n .



Sampling

- Period (s): T_s
- Frequency (Hz): $f_s = \frac{1}{T_s}$

Sample index equivalent to time:

- $t = n \cdot T_s$

Properties of signals (functions)

■ Shift:

- Shift along the horizontal axis $f_2(t) = f_1(t + \varphi)$
- Shift along the vertical axis $f_2(t) = f_1(t) + f_0$

■ Scale:

- Scale along the horizontal axis: $f_2(t) = f_1(\omega \cdot t)$
- Scale along the vertical axis: $f_2(t) = A \cdot f_1(\cdot t)$

■ Periodicity:

- $f(t)$ is periodic (with period T) if: $f(t) = f(t + T)$

■ Symmetry:

- A function is called even if: $f(-t) = f(t)$
- A function is called odd if: $f(-t) = -f(t)$

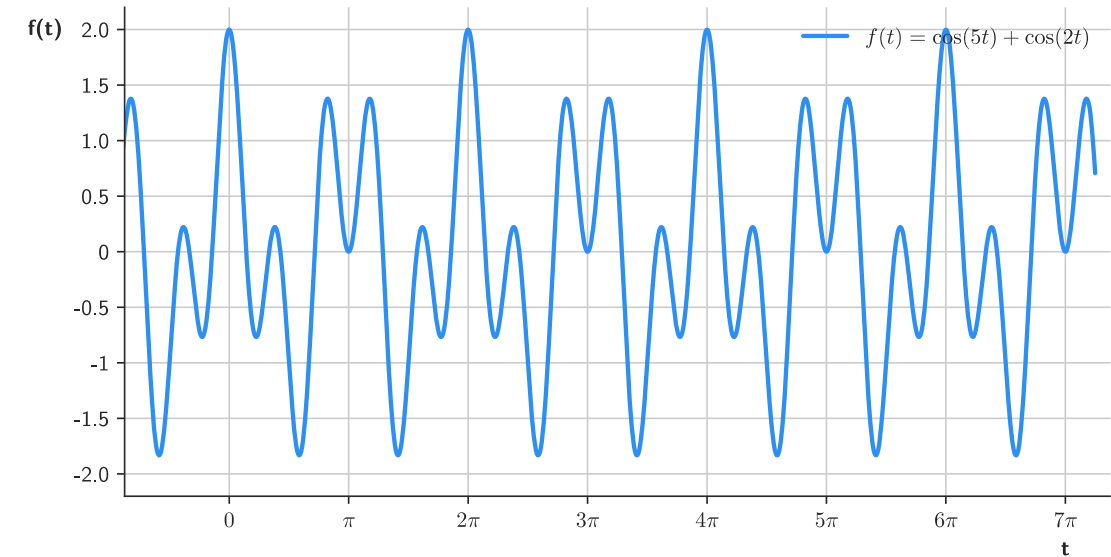
Examples (see repo):

- 01-shift-horizontal.gif
- 02-shift-vertical.gif
- 03-scale-horizontal.gif
- 04-scale-vertical
- 05-periodicity.gif
- 06-even.pdf
- 07-odd.pdf

Properties of signals (functions)

■ Combinations of functions

- Given: two functions $f_1(t)$ and $f_2(t)$
- Calculate new functions by combining them
 - Addition: $f(t) = f_1(t) + f_2(t)$
 - Subtraction: $f(t) = f_1(t) - f_2(t)$
 - Multiplication: $f(t) = f_1(t) \cdot f_2(t)$
 - Division: $f(t) = f_1(t)/f_2(t) \quad f_2(t) \neq 0$
 - Chaining: $f(t) = f_2(f_1(t))$



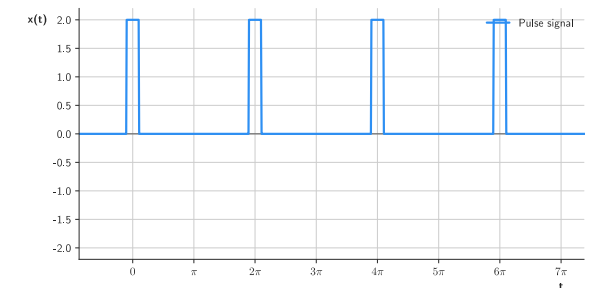
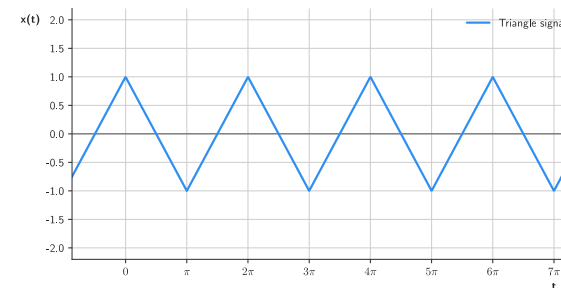
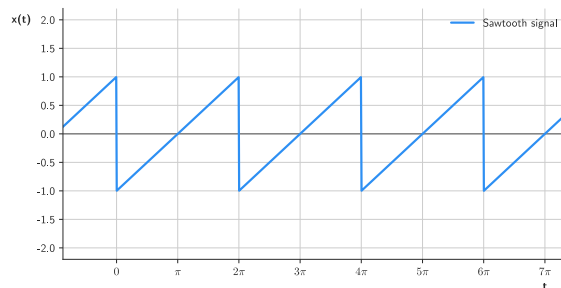
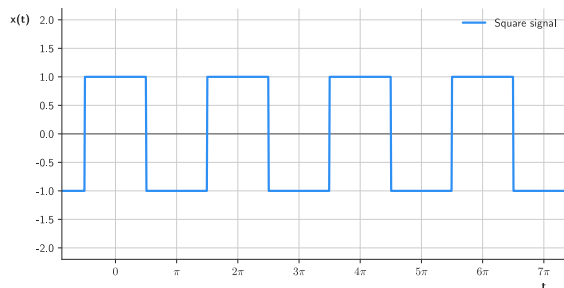
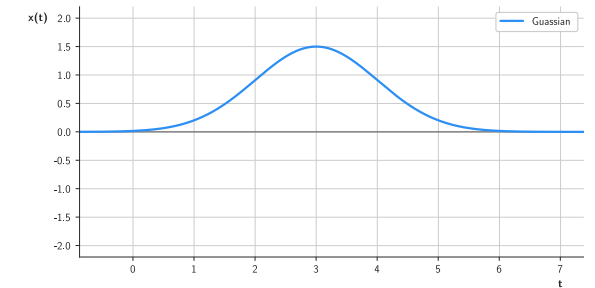
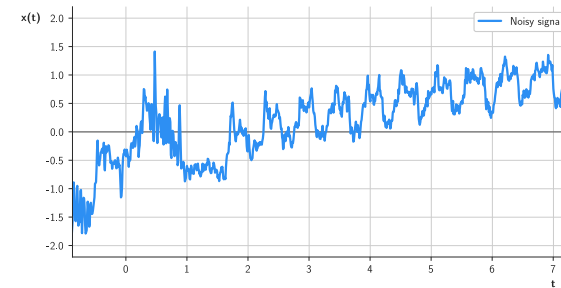
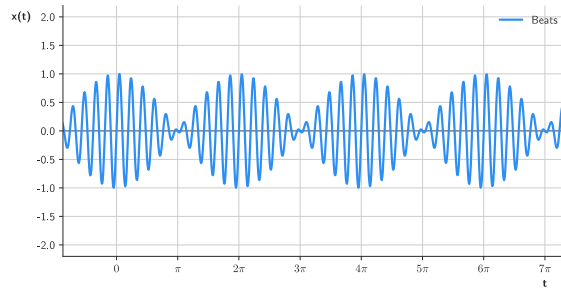
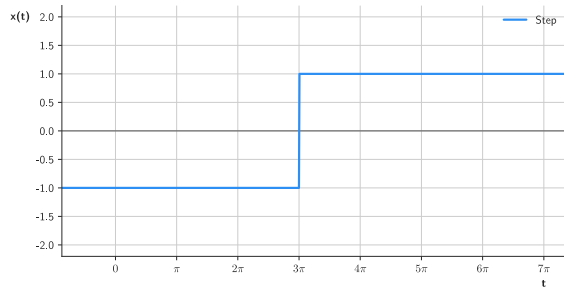
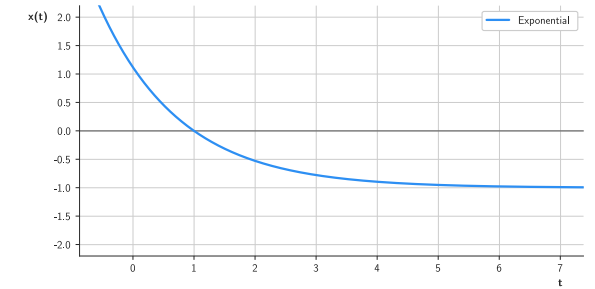
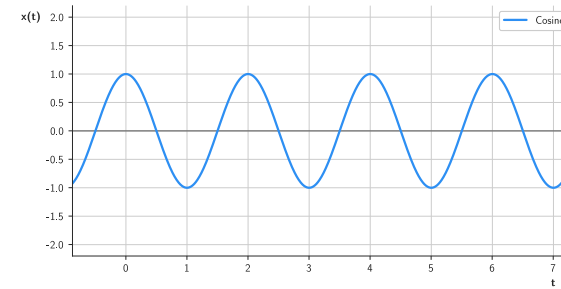
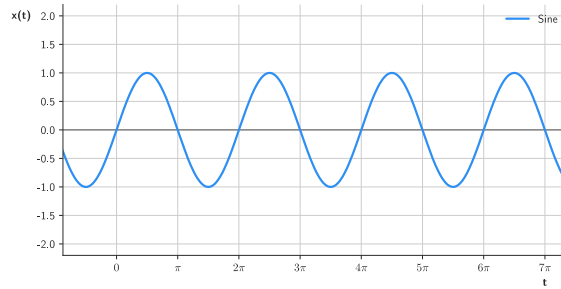
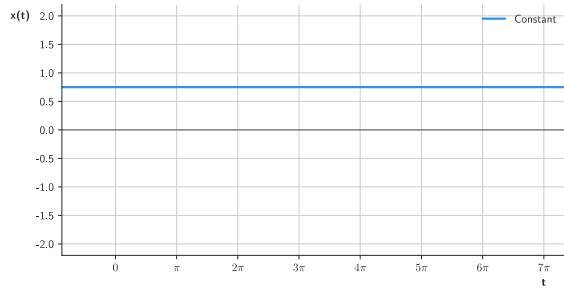
- Signals transmit energy. Using ideas from physics (or other domain knowledge), it is possible to derive the total energy of a signal as follows:

Continuous:
$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Discrete:
$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Common signal types

Task: Can you identify the function properties?
Can you even write down the function?



Common signal types

Task: Can you identify the function properties?
Can you even write down the function?

- Symmetry: even
- Periodicity: (yes)
- Function: $x(t) = a$

- Symmetry: odd
- Periodicity: 2π
- Zeros: $t = k\pi, k \in \mathbb{Z}$
- Function: $x(t) = \sin(t)$

- Symmetry: even
- Periodicity: 2π
- Zeros: $t = \frac{1}{2} + k$
- Fct.: $x(t) = \cos(2\pi t)$

- Symmetry: none
- Periodicity: none
- Fct.: $x(t) = e^{-\alpha(t-1)} - 1$
- Deriv.: strictly < 0

- Symmetry: odd, $(3\pi, 0)$
- Disc: $x(t) = \text{sgn}(\dot{t} - 3\pi)$
- Fct.: $\dot{x}(t) = 2\delta(t - 3\pi)$
- Deriv.:

- Symmetry: odd, $(0, 0)$
- Periodicity: $T_{beat} = 2\pi$
- $x(t) = \sin(\omega_1 t) + \sin(\omega_2 t)$
with $T_{beat} = \frac{2\pi}{|\omega_1 - \omega_2|}$

- Discrete-time, noisy signal without apparent symmetry
- Periodic component present for $t > 2$

- Symmetry: even
- Periodicity: none
- Shifted and scaled Gaussian ($\mu=3, \sigma=1$)

- Symmetry: even
- Periodicity: 2π
- Fct.: $x(t) = \text{sgn}(\cos(t))$
- Discontinuities at $t = \frac{k\pi}{4}$

- Symmetry: odd,
- Periodicity: 2π
- $x(t) = \frac{1}{\pi}(t \bmod 2\pi) - 1$
- Discontinuities at $t = k\pi$

- Symmetry: even
 - Periodicity: 2π
 - Value range: $[-1, 1]$
- $$x(t) = 1 - \frac{2}{\pi} ((t \bmod 2\pi) - \pi)$$

- Symmetry: even
 - Periodicity: 2π
 - Value range: $[0, 1]$
- $$x(t) = (((t + \frac{d}{2}) \bmod T) < d) - 0.5$$

Sinusoidal functions

About sines and cosines

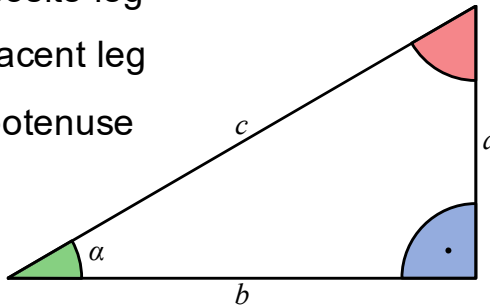
Definition and properties

1: Sine and cosine are defined as ratios of sides in a right-angled triangle.

a: opposite leg

b: adjacent leg

c: hypotenuse



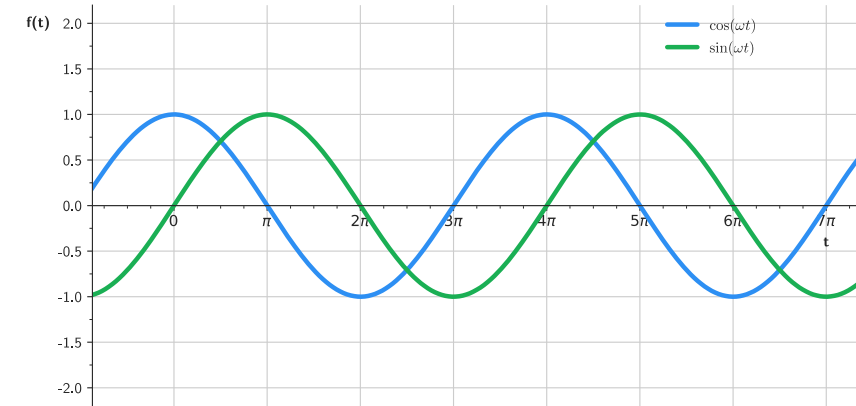
$$\sin(\alpha) = \frac{a}{c} \quad \cos(\alpha) = \frac{b}{c}$$

2: In the unit circle, the red right-angled triangle appears. Because $c=1$, $a = \sin(\alpha)$, $b = \cos(\alpha)$

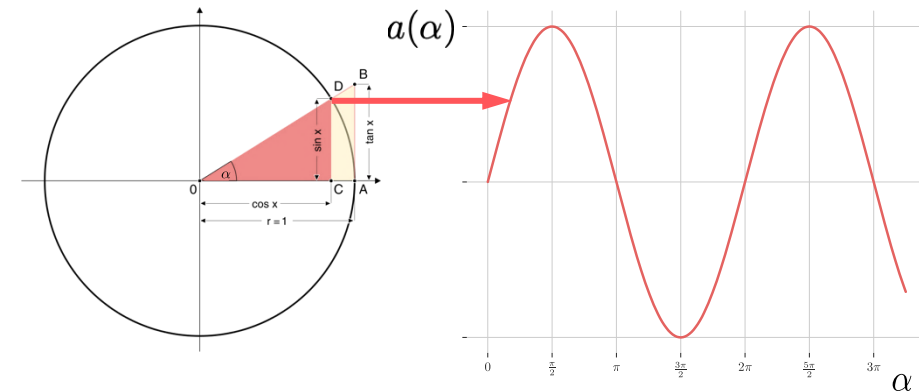
4: Introduce params for frequency, phase and amplitude to obtain the general form

$$x(t) = A \cdot \sin(\omega t + \varphi)$$

$$x(t) = A \cdot \cos(\omega t + \varphi)$$



3: By increasing the angle α , we can draw $a = \sin(\alpha)$



Calculation rules (recap)

- Trigonometric Pythagorean theorem

$$\sin^2(\omega t) + \cos^2(\omega t) = 1$$

- Cosine and sine are shifted by $\pi/2$

$$\sin(\omega t + \pi/2) = \cos(\omega t)$$

$$\cos(\omega t - \pi/2) = \sin(\omega t)$$

- Relationship between frequency and period:

- Since it must hold that $x(t) = x(t + T)$ it follows that
- f is the frequency
- ω is the angular frequency
- T is the period

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2\pi f} = \frac{1}{f}$$

Differential equations

- For the derivative of $\cos(\omega t)$, we have

$$x_1(t) = \cos(\omega t)$$

$$\frac{dx_1}{dt}(t) = \dot{x}_1(t) = -\omega \sin(\omega t)$$

$$\frac{d\dot{x}_1}{dt}(t) = \ddot{x}_1(t) = -\omega^2 \cos(\omega t)$$

- Analogously, for $\sin(\omega t)$:

$$x_2(t) = \sin(\omega t)$$

$$\ddot{x}_2(t) = -\omega^2 \sin(\omega t)$$

- This means that both $x_1(t)$ and $x_2(t)$ satisfy this differential equation:

$$\frac{1}{\omega^2} \ddot{x}(t) + x(t) = 0$$

- It can be shown that the following expression is the general solution of the differential equation above:

$$x(t) = c_1 x_1(t) + c_2 x_2(t), \quad c_1, c_2 \in \mathbb{R}$$

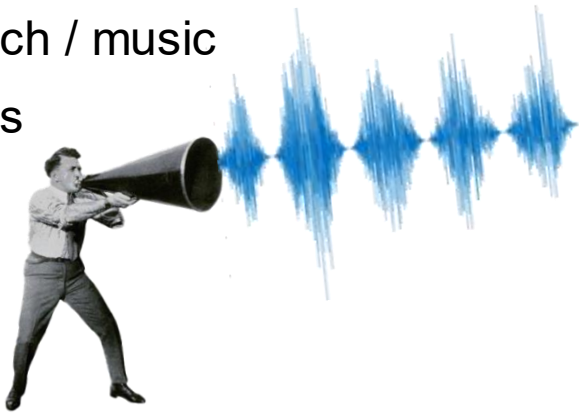
Sine waves in the real world

Physical phenomena and signals

Examples of continuous-time signals (some involving waves)

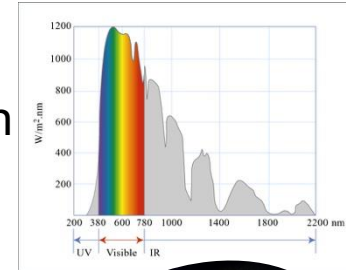
Sound waves

- Speech / music
- Sirens



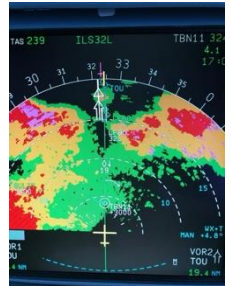
Light waves

- Video projection
- Photography
- Spectroscopy

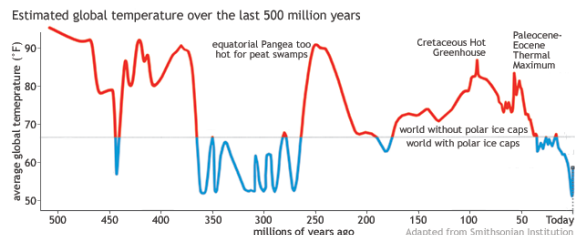


Electromagnetic waves

- Radio transmission
- Radar



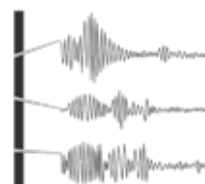
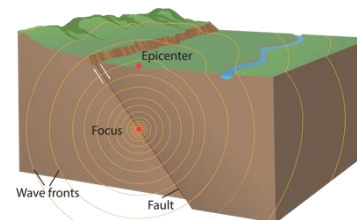
Temperature



- Global surface temperature
- Body core temperature

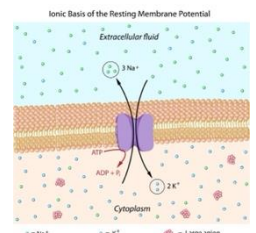
Seismic waves

- Earthquake localization
- Volcanic activity



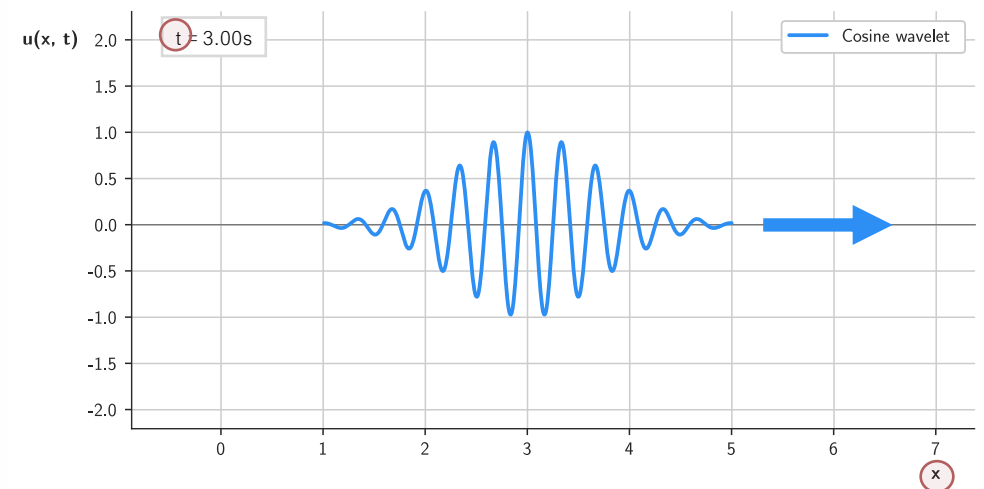
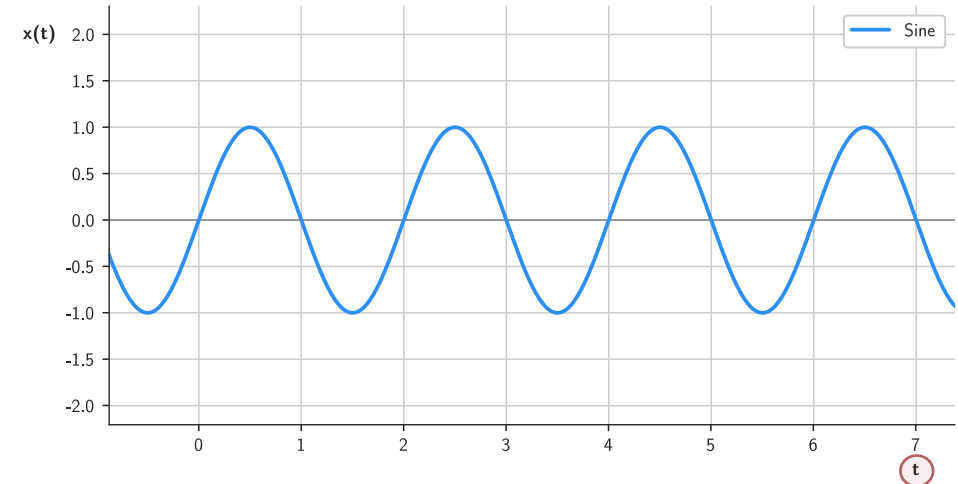
Biochemical signals

- Neural activity
- Hormones
- Cytokines



Distinction: Wave signals vs. propagating waves

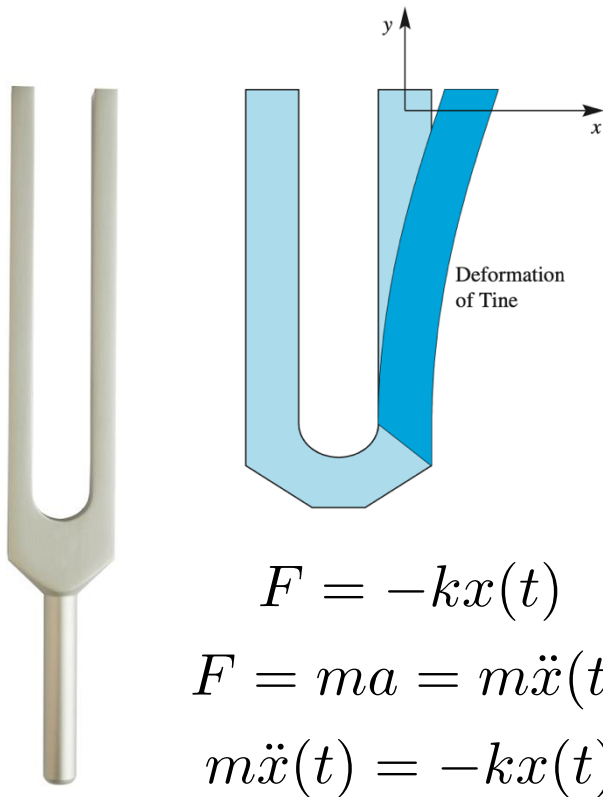
- **Signals:** $x(t)$ or $x[n]$
 - A measurement or mathematical representation of a physical quantity.
 - Purpose: Represent and encode information.
 - Defined as a function of time (cont. or discrete)
 - Do not describe spatial propagation.
- **Waves:** $u(x, t)$
 - Physical phenomena that extend in space and time
 - Disturbances that propagate through a medium or field, typically transporting energy
 - Have spatial structure and evolve according to physical laws
 - May carry information, but this is secondary to their physical nature



Tuning fork example

- Many natural phenomena can be modeled using the harmonic-oscillator differential equation introduced above.

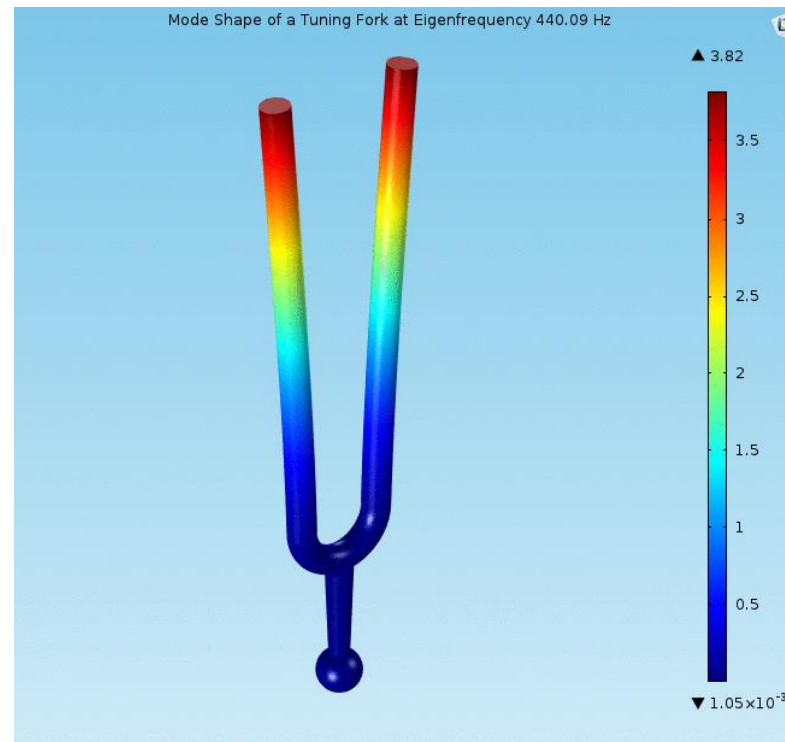
$$\frac{1}{\omega^2} \ddot{x}(t) + x(t) = 0$$



$$F = -kx(t)$$

$$F = ma = m\ddot{x}(t)$$

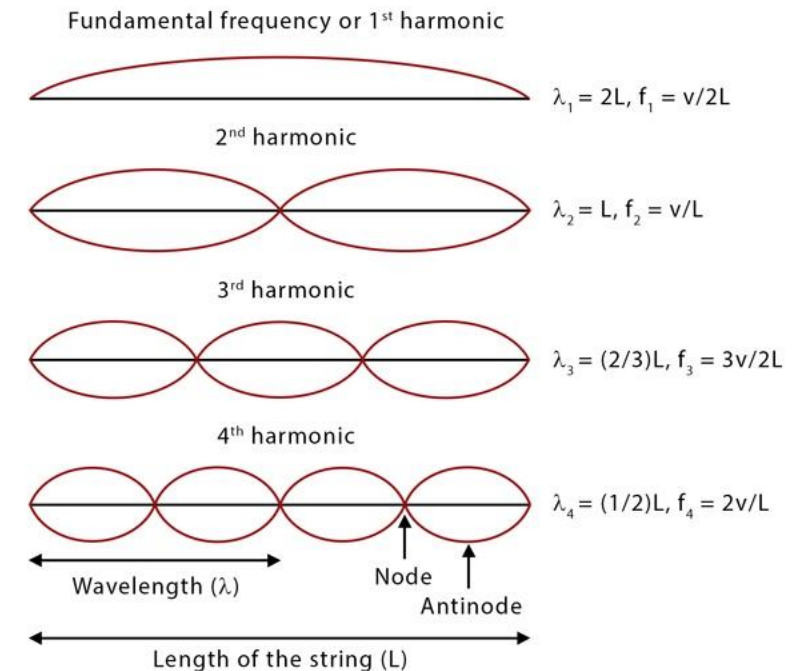
$$m\ddot{x}(t) = -kx(t)$$



The visualization on the right shows the results of an extended analysis of the same system, this time including the effects of resonance coupling. Even with this added complexity, we can still recognize that the displacement of a point on the tuning fork is well described by a simple sinusoidal motion. Image source: [Wikimedia](#)

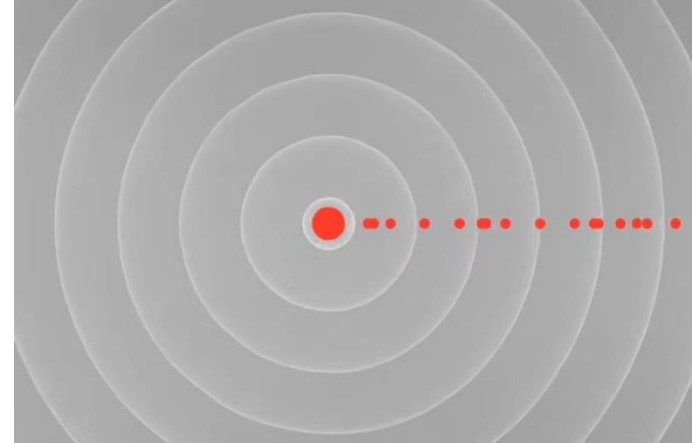
Guitar-string example

- Plucking a guitar string produces standing waves:
 - A fixed string can only vibrate in certain patterns.
 - The simplest pattern is the fundamental mode → this gives the fundamental frequency (the “note” we hear).
 - The string also vibrates simultaneously in higher modes, called overtones or harmonics.
 - Each harmonic is a whole-number multiple of the fundamental frequency.
 - The mix of fundamental + harmonics creates the characteristic sound (timbre) of a guitar.
- Even though the actual motion is complex, every point on the string moves as a sum of sinusoidal signals — the fundamental plus its harmonics.



...more about acoustic waves

- Understanding sound waves: [YouTube](#)
- How to visualize sound: [YouTube](#)



How speakers produce sound

- Basic idea: A speaker converts electrical signals into air vibrations, which our ears perceive as sound.
- Step by step:
 - Electrical signal in: The signal encodes how the air should move.
 - Coil–magnet interaction: The alternating current in the coil creates a Lorentz force, causing the coil to move back and forth.
 - Cone movement: The cone attached to the coil pushes and pulls the air, producing sound waves.
- [Explanatory video](#) (by Jacob O’Neal)

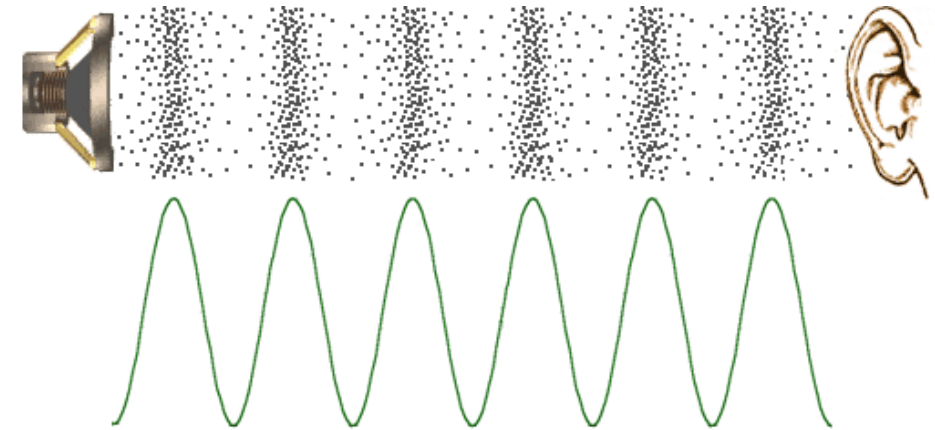


Illustration of the propagation of acoustic waves

Relationship between frequency f , speed v and wavelength λ

$$\lambda = \frac{v}{f}$$

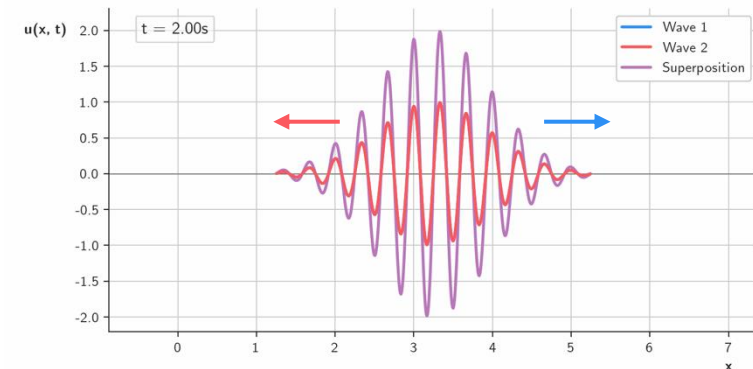
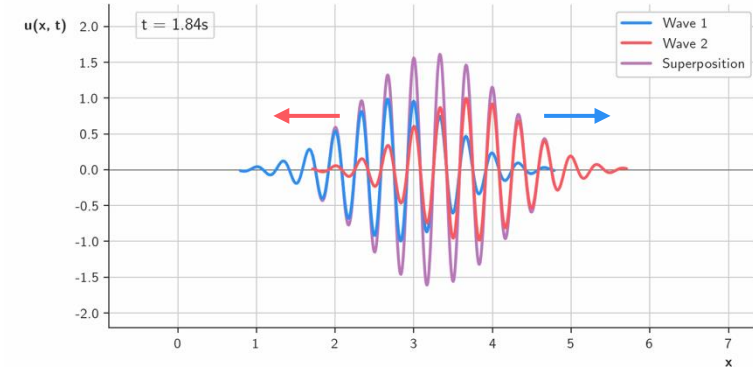
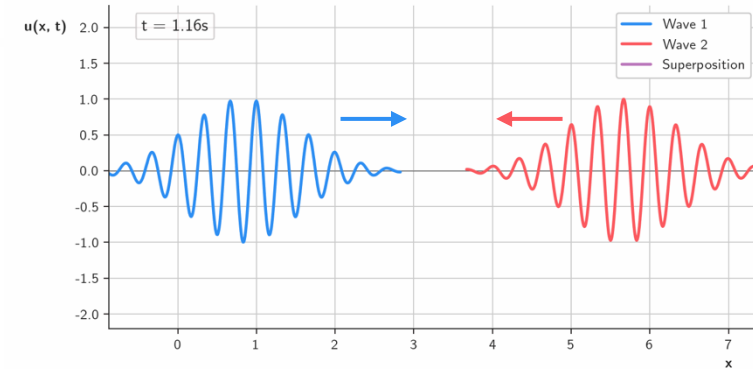
Superposition principle (of waves)

Definition: When two or more waves overlap in a medium or field, their displacements add algebraically at each point in space and time.

- Mathematically, if two waves $x_1(t)$ and $x_2(t)$ coexist at the same location, the resulting wave $x(t)$ is given by:

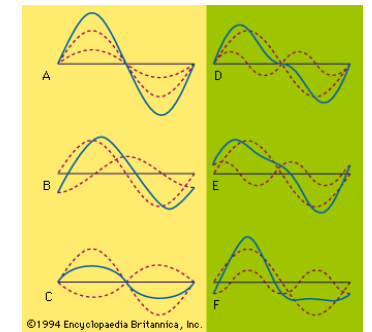
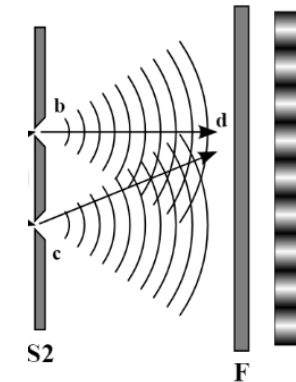
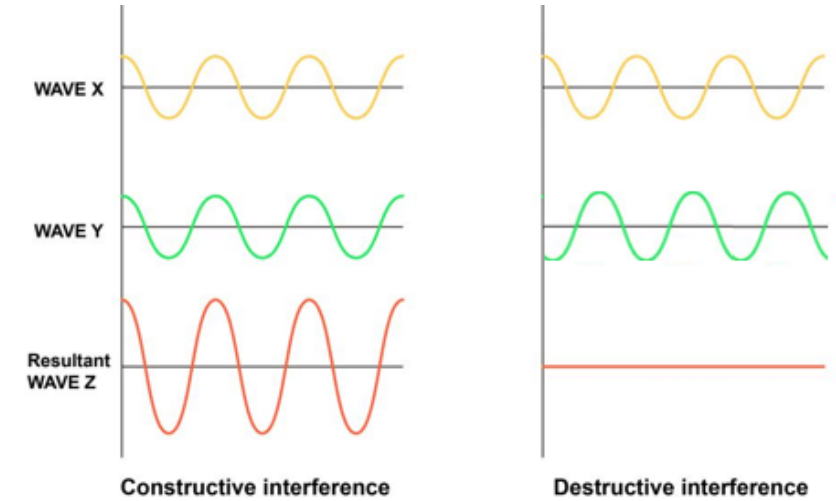
$$x(t) = x_1(t) + x_2(t)$$

- This applies to all types of waves, including sound, light, and water waves



Interference

- Wave interference occurs when two or more waves overlap and combine to form a new wave pattern.
- Constructive interference:** The Waves are in phase (peaks align with peaks, troughs with troughs), producing a wave with increased amplitude.
- Destructive interference:** Waves are out of phase (peak meets trough), causing partial or complete cancellation.
- Videos:
 - [Wave interference](#)
 - [Wave interference for two sound sources](#)



Left: Interference pattern from two light sources. **Right:** Partial interference with differing amplitudes or wavelengths.

Interference: Beating signals

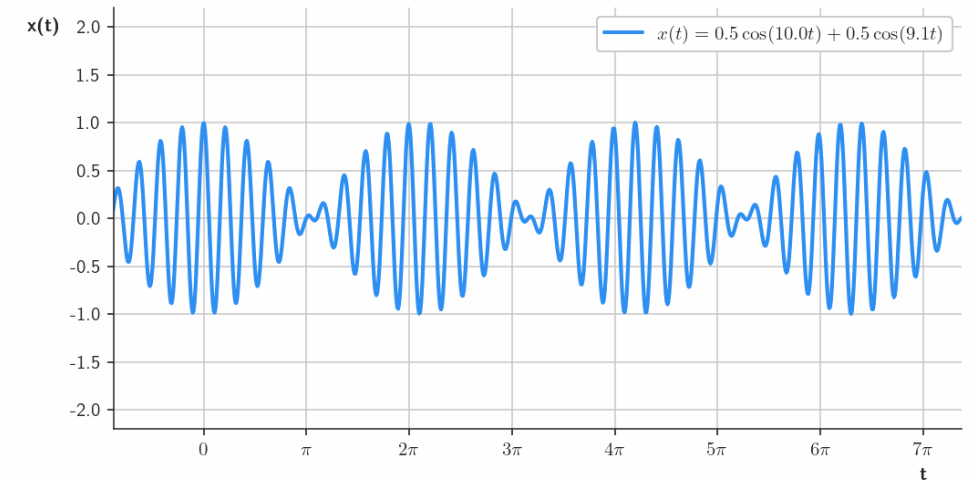
- A beat is an interference pattern that occurs when two waves of slightly different frequencies overlap:

$$x(t) = a_1 \cos(2\pi f_1 t) + a_2 \cos(2\pi f_2 t)$$

- Beats are perceived as periodic variations in amplitude (pulsating sound)
- One can show that the beat frequency of the beat is equal to the difference in frequency between the two original sounds.

$$f_{beat} = |f_1 - f_2|$$

- Video: [Beating effect demonstrated with tuning forks](#)



Animation: Superposition of two cosine waves, where the second wave gradually increases its frequency over time.

Superposition principle (linear, time-invariant systems)

■ Context: Modelling (signal-generating) systems

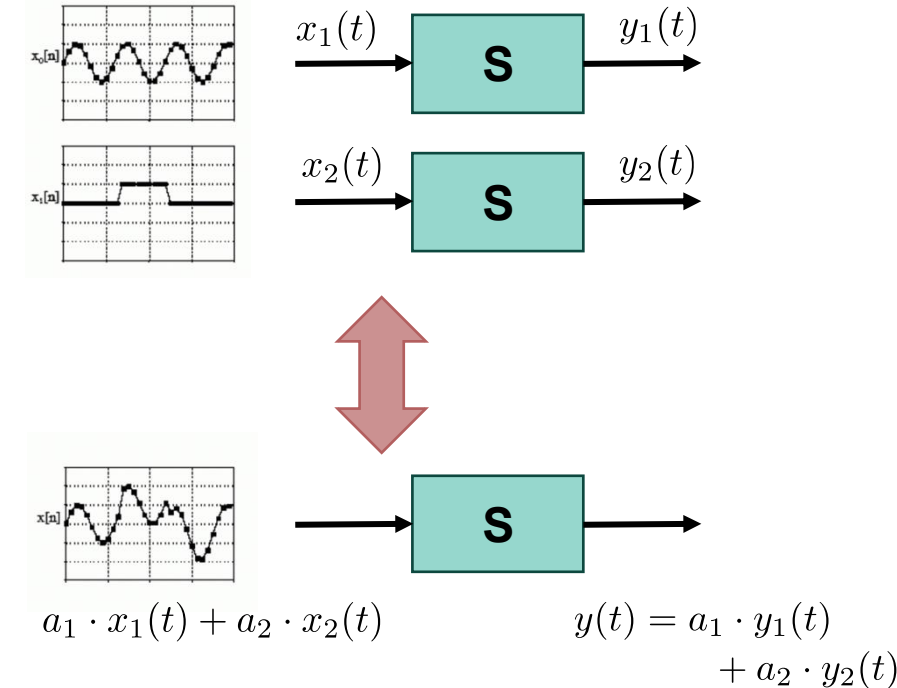
- Dynamic systems are often described by differential equations. (We have already seen an example).
- Goal: Model the input-output behavior of a system.

■ Superposition principle:

- A system S obeys the superposition principle if its response to a sum of inputs equals the sum of the responses to each input.
- Systems with this property are called linear systems.

■ Related: Linear functions

- A function is called linear if it satisfies
- Additivity: $f(x_1 + x_2) = f(x_1) + f(x_2)$
- Homogeneity: $f(\alpha x) = \alpha f(x)$



You learned about **two** superposition principles: one for waves and one for LTI systems.